

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. CHB-502: Inorganic Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Display the effect of a catalyst in a thermodynamically favored chemical reaction by drawing energy profile diagrams.
- (b) Which statement about EDTA complexes of Mg^{2+} and Ca^{2+} is correct? (i) EDTA complex of Ca^{2+} is more stable than Mg^{2+} , or (ii) EDTA complex of Mg^{2+} is more stable than Ca^{2+} . Give reasons.
- (c) Arrange the following ligands in order of their increasing field strength: ox^{2-} , OH^- , H_2O , py, en, NCS^- .
- (d) Draw and describe the structural features of $\text{Li}_4(\text{CH}_3)_4$.
- (e) Draw the structure of $\text{Ti}(\eta^1\text{-C}_5\text{H}_5)_2(\eta^5\text{-C}_5\text{H}_5)_2$ and write its IUPAC name.

Question 2.

- (a) How is the Laporte Selection Rule relaxed in $[\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2+}$ and $[\text{CoCl}_4]^{2-}$ complexes? [6]
- (b) Predict the empirical formula of a metal complex using the following data: $k_3/k_2 = 3/8$ and $k_4/k_3 = 9/16$, where k_n denotes stepwise formation constants. [8]

Question 3.

- (a) Explain 'trans-effect' with proper reactions. Write down the electrostatic polarization theory and π -bonding theory of the trans-effect. [8]
- (b) In the electronic spectrum of $[\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2+}$, assign the type of transitions to the bands at 50000 cm^{-1} , 25000 cm^{-1} , and 17000 cm^{-1} . [6]

Question 4.

- (a) Write the products of the following organometallic reactions: [8]
 - (i) $\text{R}_3\text{SnH} + \text{Me-C} \equiv \text{C-Me} \rightarrow ?$
 - (ii) $\text{Fe}(\text{CO})_5 + \text{C}_5\text{H}_6 \rightarrow ?$

- (b) The enthalpy of hydration (ΔH_{hyd}) of Cr^{2+} is -460 kcal/mol . In the presence of CFSE, discuss how hydration energy is affected by crystal field effects. [6]
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Question 5.

- (a) Differentiate between thermodynamic stability and kinetic stability of coordination complexes. [6]
- (b) In the electronic absorption spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, along with the main absorption band at 20300 cm^{-1} , a shoulder band is also observed. Explain why. [8]